

Homework 1

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Problem 1

a)

$$1. Z_C = \sqrt{Z_0 Z_L} = 70.71\Omega$$

$$2. Z_C = \sqrt{Z_0 Z_L} = 100\Omega$$

b)

At the center frequency f_0 , the quarter-wave transformer satisfies

$$\ell = \frac{\lambda_0}{4}$$

where

$$\lambda_0 = \frac{v_p}{f_0}$$

Hence,

$$\ell = \frac{v_p}{4f_0}$$

At an arbitrary frequency f , the wave number is

$$k(f) = \frac{2\pi}{\lambda(f)} = \frac{2\pi f}{v_p}$$

Therefore,

$$k(f)\ell = \frac{2\pi f}{v_p} \cdot \frac{v_p}{4f_0} = \frac{\pi f}{2f_0}$$

and

$$2k(f)\ell = \pi \frac{f}{f_0}$$

As is learned from our course, the reflection coefficient at the load is

$$\Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C}$$

the input reflection coefficient is

$$\Gamma_{in}(f) = \Gamma_L e^{-2jk(f)\ell}$$

substituting

$$\Gamma_{in}(f) = \Gamma_L e^{-j\pi f/f_0}$$

thus the input impedance is

$$Z_{in}(f) = Z_C \frac{1 + \Gamma_{in}(f)}{1 - \Gamma_{in}(f)}$$

The required quantity is the input reflection coefficient with respect to the system reference impedance Z_0 :

$$S_{11}(f) = \frac{Z_{in}(f) - Z_0}{Z_{in}(f) + Z_0}$$

Now we combine the results above:

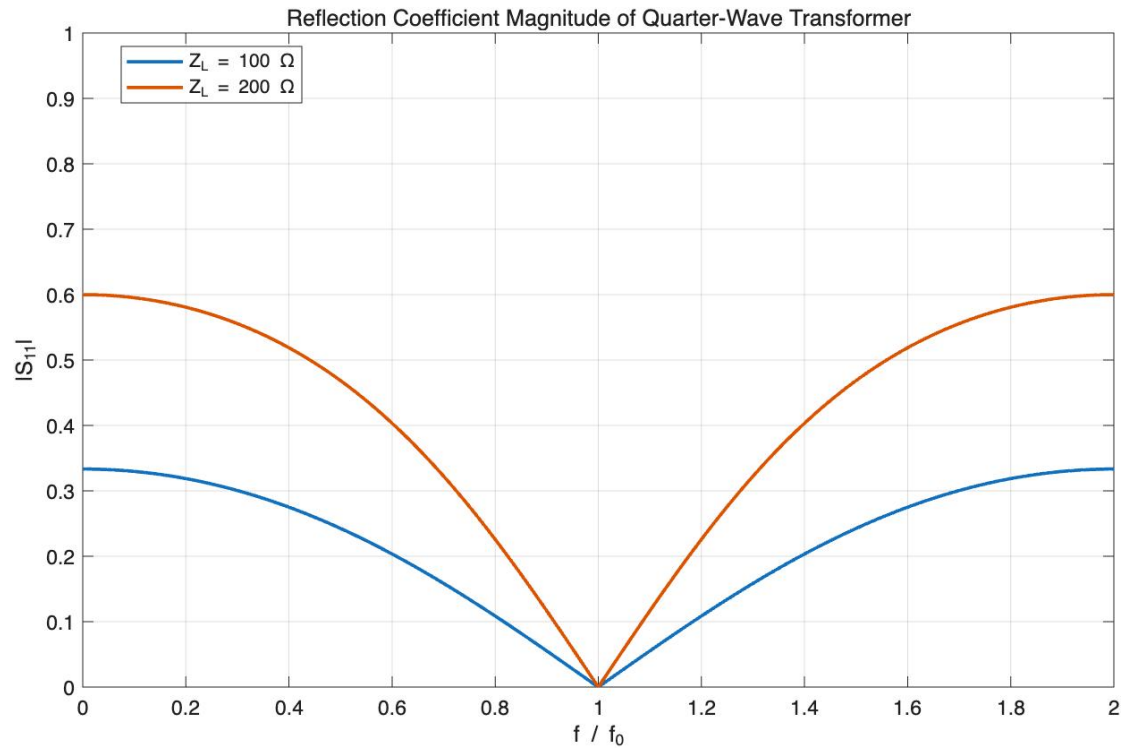
$$S_{11}(f) = \frac{Z_C \frac{1 + \Gamma_L e^{-j\pi f/f_0}}{1 - \Gamma_L e^{-j\pi f/f_0}} - Z_0}{Z_C \frac{1 + \Gamma_L e^{-j\pi f/f_0}}{1 - \Gamma_L e^{-j\pi f/f_0}} + Z_0}$$

where

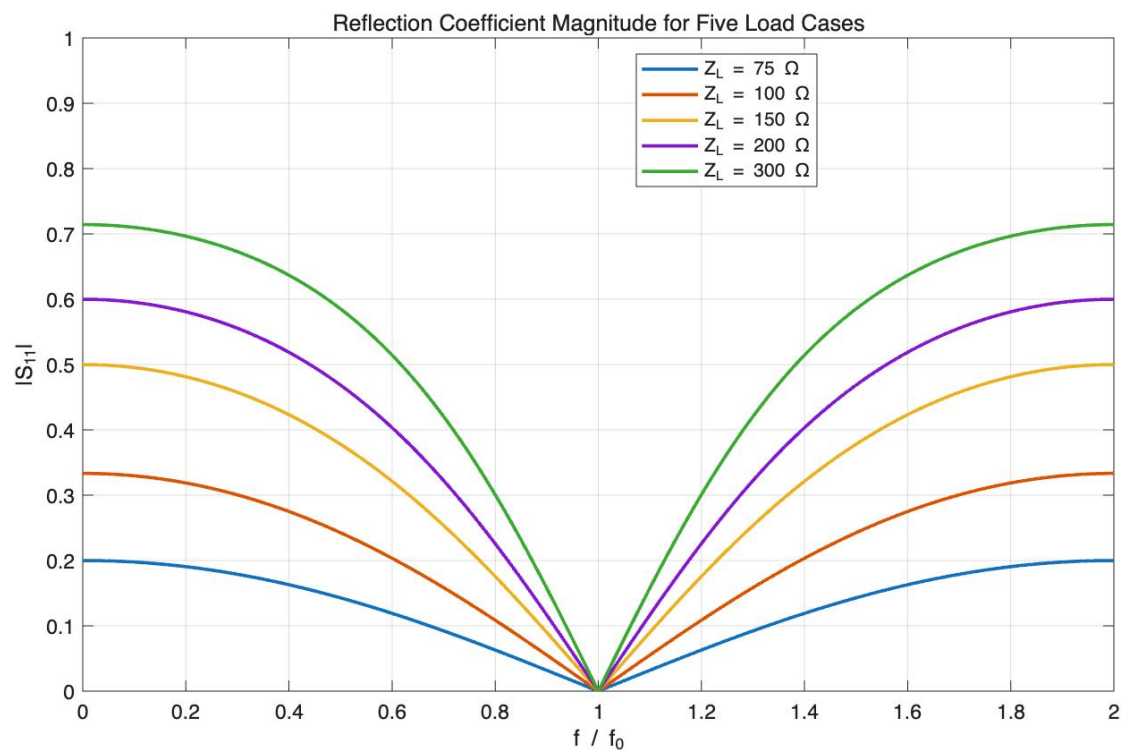
$$\Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C}$$

$$Z_C = \sqrt{Z_0 Z_L}$$

Then we use matlab to plot the figure:



We add three more different values of Z_L to see the results:



At $\frac{f}{f_0} = 1$, all curves go to zero. This means the QWT provides perfect matching at the design frequency f_0 . At this frequency no reflection occurs.

Away from f_0 , $|S_{11}|$ increases. This means the matching becomes worse when the frequency moves away from the center frequency. Therefore, the QWT is frequency-dependent and works best only around f_0 .

What's more, as Z_L becomes larger, the curve becomes higher. In other words, a larger difference between Z_L and Z_0 produces stronger reflection away from f_0 .

For example, $Z_L = 300\Omega$ gives the largest $|S_{11}|$, while $Z_L = 75\Omega$ gives the smallest $|S_{11}|$. This also means that when the load is closer to $Z_0 = 50\Omega$, the matching is better over a wider frequency range. When the load is much larger than 50Ω , the useful matching bandwidth becomes narrower.

Problem 2

According to the problem

$$\ell = \frac{\lambda_0}{2} \text{ at } f_0$$

And as we learned from the course

$$\Gamma_L = \frac{Z_0 - Z_C}{Z_0 + Z_C}$$

$$\Gamma_{in}(f) = \Gamma_L e^{-2jk(f)\ell}$$

$$Z_{in}(f) = Z_C \frac{1 + \Gamma_{in}(f)}{1 - \Gamma_{in}(f)}$$

with

$$k(f)\ell = \frac{2\pi f}{v_p} \cdot \frac{v_p}{2f_0} = \pi \frac{f}{f_0}$$

and the input reflection coefficient is

$$S_{11}(f) = \frac{Z_{in}(f) - Z_0}{Z_{in}(f) + Z_0}$$

For a lossless two-port network, the scattering matrix is unitary. Therefore, the following relation holds:

$$|S_{11}(f)|^2 + |S_{21}(f)|^2 = 1$$

Hence, the transmission coefficient magnitude can be obtained from the reflection coefficient magnitude:

$$|S_{21}(f)| = \sqrt{1 - |S_{11}(f)|^2}$$

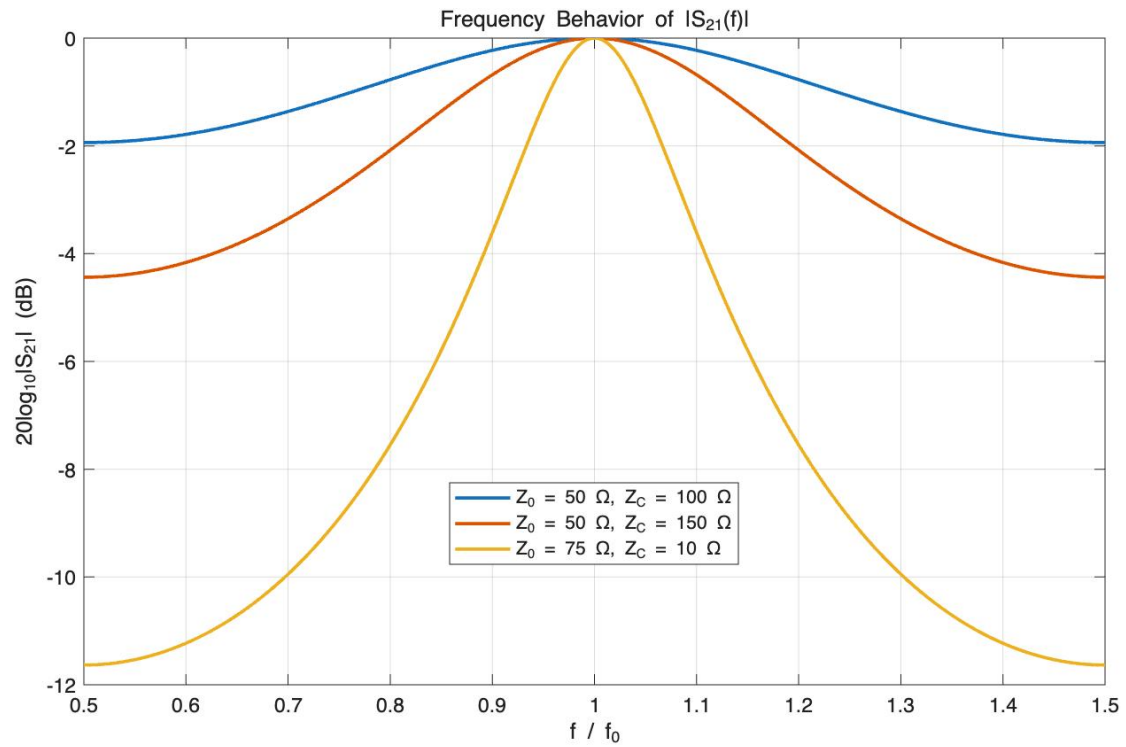
Thus,

$$|S_{21}(f)| = \sqrt{1 - \left| \frac{Z_{in}(f) - Z_0}{Z_{in}(f) + Z_0} \right|^2}$$

The required quantity in dB is

$$20 \log_{10} |S_{21}(f)|$$

We then plot the figure in matlab:



As is depicted from the figure, all three curves reach 0 dB at $\frac{f}{f_0} = 1$. This means $|S_{21}(f)|=1$ at the center frequency, so the transmission is maximum at f_0 . When the frequency moves away from f_0 , the transmission decreases. This shows that the half-wave transmission line gives the best transmission only at the design frequency.

The three cases are different because the mismatch between Z_0 and Z_C is different. A larger mismatch gives a lower $|S_{21}(f)|$ away from f_0 . For this reason, the case $Z_0=75\Omega$, $Z_C=10\Omega$ has the worst transmission, while $Z_0=50\Omega$, $Z_C=100\Omega$ has the best one among the three cases.

In summary, the figure shows that the transmission is perfect at f_0 , but it becomes worse away from f_0 , especially when the impedance mismatch is large.